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Salem (Dt)

COMMON QUARTERLY EXAMINATION - 2025

CLASS:12 PHYSICS TOTAL: 70

TIME : 3 HrS

I CHOOSE THE BEST ANSWER : 15X1=15

1. Which charge configuration produces a uniform electric field?
   a) point Charge b) infinite uniform line charge
   c) uniformly charged spherical shell d) uniformly charged infinite plane
2. An electric dipole is placed at an alignment angle of 30° with an electric field of $2 \times 10^2\ NC^{-1}$. It experiences torque equal to 8 Nm. The charge on the dipole if the dipole length is 1 cm is
   a) 4 mC b) 7 mC c) 5 mC d) 8 mC
3. A toaster operating at 120 V has a resistance of 240 Ω The power is
   a) 400 W b) 60 W c) 480 W d) 240 W
4. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is
   a) 0.20Ω b) 0.5 Ω c) 0.8 Ω d) 1.0 Ω
5. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is
   a) 1.2 Am^2 b) 10 Am^2 c) 0.5 Am^2 d) 0.8 Am^2
6. A non-conducting charged ring of charge q, mass m and radius r is rotated with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2r}$ d) $\frac{q}{2m}$
7. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is
   a) 2 A b) 18 A c) 12 A d) 1 A
8. In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current in the circuit is
   a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{2}$ d) Zero
9. Which of the following electromagnetic radiation is used for viewing objects through fog
   a) microwave b) infrared c) X- rays d) gamma rays
10. For light incident from air on a slab of refractive index 2. Maximum possible angle of refraction is,
    a) 45^0 b) 30^0 c) 60^0 d) 90^0
11. Stars twinkle due to,
    a) reflection b) total internal reflection
    c) refraction d) polarization
12. The electricfield that can balance a charged particle of mass 3.2 X 10^{-6} kg is (Given that the charge on the particle is 1.6 X 10^{-19} C)
    a) 19.6 X 10^{-8} NC^{-1} b) 20 X 10^{-6} NC^{-1} C) 19.6 X 10^{8} NC^{-1} d) 20 X 10^{6} NC^{-1}
13. P,Q,R and S are four resistances of 2Ω,2Ω,2Ω and 3Ω respectively. They are used to form a Wheatstone bridge. The resistance S is shunted with a resistances X in order to get the bridge balanced. The value of X will be
    a) 4 Ω b) 6 Ω C) 8 Ω d) 3 Ω
14. A 60 μF capacitor is connected to a 110 V (rms), 60 Hz ac supply. The rms value of current in the circuit is
    a) 1.49 A b) 14.9 A C) 2.49 A d) 24.9
15. A radio can tune to any station in 7.5 MHz to 12 MHz band . The corresponding wavelength band is
    a) 40m to 25m b) 30m to 25 m C) 2m to 10m d) 10m to 5m

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II. ANSWER ANY SIX QUESTIONS : Q.No. 24 COMPULSORY: 6X2=12

16. Mention the ways of producing induced emf
17. What is corona discharge? (or) action of point
18. Define current density.Give its units
19. What is meant by hysteresis?
20. Resistance of a material at 10 °C and 40 °C are 45 Ω and 85 Ω respectively. Find its temperature coefficient of resistivity.
21. What is displacement current?
22. Write the use of capacitor in computer key board?
23. Why does sky appear blue?
24. For a transparent medium relative permeability and permittivity μ_r and ϵ_r are 1.0 and 1.44respectively. then what is the velocity of the light in a medium ?.

III. ANSWER ANY SIX QUESTIONS. Q.No. 33 COMPULSORY: 6X3=18

25. Discuss the conversion of galvanometer in to a voltmeter.
26. What are Fraunhofer lines? How are they useful in the identification of elements present in the Sun?
27. a) Define electric flux
    b) Calculate the electric flux through the rectangle of sides 5 cm and 10 cm kept in the region of a uniform electric field 100 NC^{-1} with the angle θ is 60°
28. Explain the various energy losses in a transformer.
29. Obtain the expression for a capacitance for a parallel plate capacitor.
30. Explain the determination of the internal resistance of a cell using voltmeter.
31. Derive the equation for effective focal length for lenses in contact.
32. A short bar magnet has a magnetic moment of 0.5 JT^{-1}. Calculate magnitude and direction of the magnetic field produced by the bar magnet which is kept at a distance of 0.1 m from the centre of the bar magnet along (a) axial line of the bar magnet and (b) normal bisector of the bar magnet.
33. A 500 μH inductor, $\frac{80}{\pi^2}$ pF capacitor and a 628 Ω resistor are connected to form a series *RLC* circuit. Calculate the resonant frequency and Q-factor of this circuit at resonance.

IV. ANSWER ALL THE QUESTIONS: 5X5=25

34. (A) Obtain the condition for bridge balance in Wheatstone'bridge. ***(OR)***
    (B) Derive the mirror equation and the equation for lateral magnification.
35. (A) Discuss the working Of cyclotron in detail. ***(OR)***
    (B) What is absorption spectrum? Explains its types.
36. (A) (i)What is optical path
    (ii) Light travels from air in to a glass slab of thickness 50 cm and refractive index 1.5.
        (a) What is the speed of light in the glass slab?
        (b) What is the time taken by the light to travel through the glass slab?
        (c) What is the optical path of the glass slab? ***(OR)***
    (B) (i) Define RMS value alternating current.
    (ii)The equation for an alternating current is given by $i = 77 \sin 314t$.
    Find the peak current, frequency, time period and instantaneous value of current at $t = 2$ ms.
37. (A) Explain in detail the principle, construction and working of a Vande Graaff Generator. ***(OR)***
    (B) Explain cells are connected in series.
38. (A) Calculate the electricfield due to a dipole on its axialline. ***(OR)***
    (B) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.

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